TEXORA · LECTURE SHEET

Pollution and Its Effects on the Living World

Pollution and pollutants defined and classified; the structure of the atmosphere; and the three great air phenomena — the greenhouse effect, photochemical smog and ozone-layer depletion — through which pollution reaches every living thing.

TE-4204 · ENVIRONMENTAL ENGINEERING AND POLLUTION CONTROL · TOPIC 2 · LEVEL 4 TERM 2

1. Pollution and pollutants

Pollution is the introduction by human activity of substances or energy (heat, noise, light, radiation) into the environment at rates or concentrations that harm living organisms, damage materials or ecosystems, or interfere with legitimate uses of air, water and land. A **pollutant** is the substance or energy form responsible. The same chemical can be a resource in one place and a pollutant in another: phosphate fertilises a field but eutrophies a lake, and carbon dioxide feeds plants yet, in excess, warms the climate. Pollution is therefore about *concentration, location and rate*, not about "good" and "bad" chemicals.

Pollutants are classified in several useful ways. **Primary pollutants** are emitted directly (soot, sulphur dioxide, carbon monoxide, dyehouse effluent); **secondary pollutants** form in the environment by reaction (ozone and photochemical smog from nitrogen oxides and hydrocarbons; acid rain from sulphur and nitrogen oxides). By source behaviour, a **point source** discharges at an identifiable place (a factory outlet, a chimney) and can be metered and licensed, while a **non-point (diffuse) source** — farm run-off, city storm water — arrives over a wide area and is much harder to control. By persistence, **biodegradable** pollutants (sewage, starch size) are broken down by micro-organisms if the load is within the water body's assimilative capacity, while **non-biodegradable** ones (many synthetic chemicals, heavy metals, plastics) accumulate indefinitely. Energy pollutants — noise, waste heat and light — leave no residue but act while they are emitted.

2. The atmosphere and its structure

Almost all air pollution acts in the lowest layer of the atmosphere, so its structure matters. The atmosphere thins with altitude and is divided by its temperature profile into layers:

LAYER	APPROXIMATE HEIGHT	TEMPERATURE BEHAVIOUR	SIGNIFICANCE
Troposphere	0–12 km	Falls with height (about 6.5 °C per km)	Weather, clouds; nearly all pollution is emitted and dispersed here
Stratosphere	12–50 km	Rises with height (ozone absorbs UV)	Contains the ozone layer; very stable, pollutants persist for years
Mesosphere	50–85 km	Falls to the coldest temperatures in the system	Meteors burn up here
Thermosphere	85–600 km	Rises strongly (solar absorption)	Auroras; radio-reflecting ionosphere lies within it
Exosphere	Above ~600 km	—	Merges into space; satellites orbit here

Two practical consequences follow. First, the troposphere's falling temperature normally mixes air vertically and dilutes emissions — but when a warm layer overlies cool air (a **temperature inversion**), mixing stops and pollutants accumulate at ground level, producing pollution episodes. Second, the boundary between the turbulent troposphere and the stable stratosphere is hard to cross, so substances that do reach the stratosphere — chlorofluorocarbons are the notorious example — stay there for decades, which is what makes ozone depletion a long-term problem.

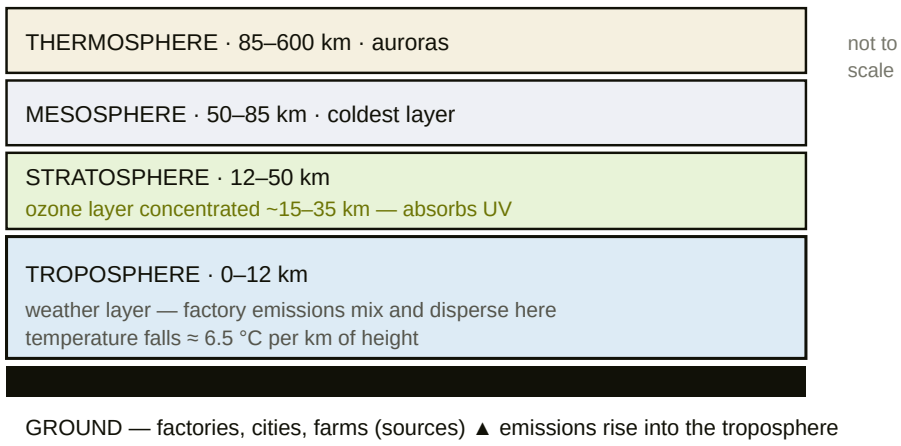


Figure 1 — The layers of the atmosphere. Heights and features are as labelled; the drawing is schematic (band thicknesses are not proportional to real heights). Pollution control in this course concerns the troposphere; ozone depletion concerns the stratosphere.

3. Types of pollution and their reach

TYPE	TYPICAL POLLUTANTS	EFFECTS ON THE LIVING WORLD
Air	Particulates, SO_2, NO_x, CO, ozone, VOCs	Respiratory disease, acid rain, crop damage, climate forcing
Water	BOD/COD load, colour, salts, metals, nutrients	Oxygen depletion, fish kills, eutrophication, disease
Soil / land	Heavy metals, persistent chemicals, sludge, plastics	Loss of soil fertility, contaminated crops and groundwater
Noise	Machinery, transport, generators	Hearing loss, stress, sleep and communication disturbance
Thermal	Hot water and air discharges	Lower dissolved oxygen, shifted aquatic communities
Radioactive	Radionuclides from industry, medicine, accidents	Long-lived contamination, genetic damage

Effects scale from the cell to the planet. A pollutant may act **directly** (soot irritating a lung), **indirectly** (acid rain acidifying a lake that then releases toxic aluminium), by **accumulation** (metals concentrating up a food chain — biomagnification), or **synergistically**, where two pollutants together do more damage than the sum of their separate effects, as with the components of photochemical smog.

4. The greenhouse effect

The **greenhouse effect** is the natural process that keeps the Earth habitable. Short-wave solar radiation passes largely through the atmosphere and warms the surface; the warm surface re-radiates long-wave (infrared) energy; certain gases — water vapour, carbon dioxide, methane, nitrous oxide and others — absorb part of this infrared radiation and re-emit it in all directions, including back downward. Without this blanket the average surface temperature would be far below freezing. The environmental problem is the **enhanced** greenhouse effect: human activity is increasing the concentrations of these gases, thickening the blanket and warming the climate system.

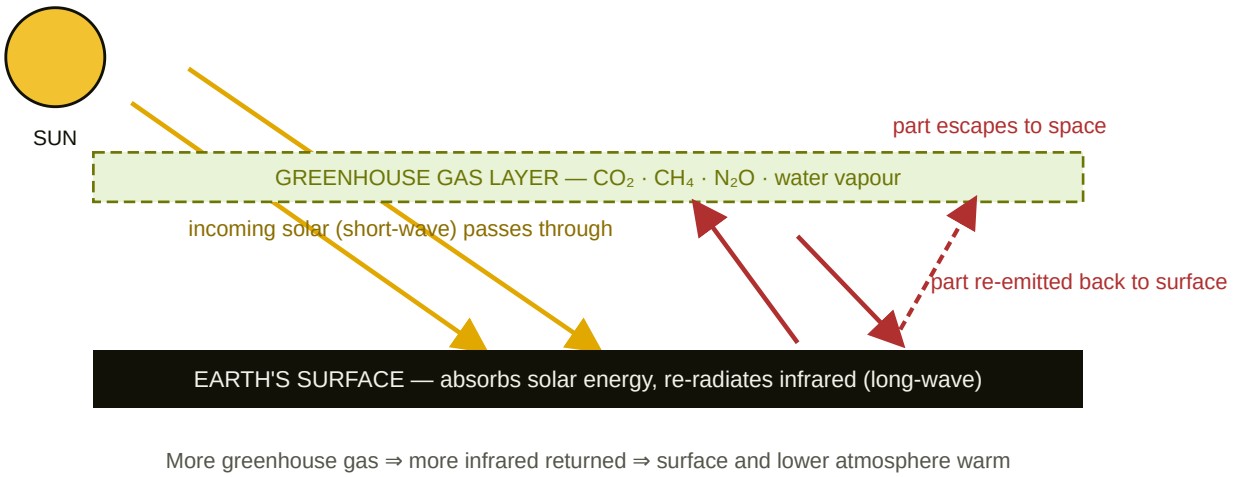


Figure 2 — The greenhouse effect. Solar short-wave radiation reaches the surface; the surface's long-wave radiation is partly absorbed by greenhouse gases and re-emitted in all directions. Human emissions strengthen this natural effect.

Gases differ greatly in potency, expressed as **global warming potential (GWP)** over a chosen horizon (usually 100 years), relative to CO_2 = 1:

GAS	MAIN HUMAN SOURCES	APPROXIMATE GWP (100-YEAR)
Carbon dioxide (CO_2)	Fuel combustion, deforestation	1
Methane (CH_4)	Livestock, rice fields, landfills, gas leakage	≈ 28
Nitrous oxide (N_2O)	Fertilisers, some chemical processes	≈ 265
Hydrofluorocarbons (HFCs)	Refrigerants, foams	Hundreds to thousands, by compound

WORKED EXAMPLE — CO_2-EQUIVALENT EMISSIONS

A boiler house reports annual emissions of 12,000 tonnes of CO_2 and 50 tonnes of methane (from an associated biogas/landfill source). Express the total in CO_2-equivalent using GWP(CH_4) ≈ 28.
Methane contribution = 50 × 28 = **1,400 tonnes CO_2e**.
Total = 12,000 + 1,400 = **13,400 tonnes CO_2e per year**.
Lesson: a small mass of a potent gas can rival a large mass of CO_2 — which is why inventories are kept in CO_2e, not in tonnes of gas.

5. Photochemical smog

Photochemical smog is a secondary pollution mixture produced when nitrogen oxides (NO_x) and volatile organic compounds (VOCs) react in strong sunlight. The chemistry centres on ozone: sunlight splits NO_2, freeing an oxygen atom that forms ozone (O_3); normally NO destroys ozone again, but VOCs interrupt the cycle so ozone accumulates, together with other oxidants such as peroxyacetyl nitrate (PAN). The visible result is a brownish haze; the health result is eye and throat irritation and worsened asthma; the agricultural result is leaf injury and reduced yields. It is worst on hot, still, sunny days in vehicle-dense cities — the classic Los Angeles pattern — and it contrasts with the older **London-type (sulphurous) smog**, a winter mixture of smoke, fog and SO_2 from coal burning. Control follows from the chemistry: cut either precursor — combustion NO_x (burner tuning, exhaust after-treatment) or VOCs (solvent management, vapour recovery) — and ozone formation falls.

6. Ozone-layer depletion

High in the stratosphere, ozone forms a dilute but vital shield absorbing most of the Sun's harmful ultraviolet (UV-B) radiation. **Chlorofluorocarbons (CFCs)** — once common in refrigeration, aerosols and foam blowing — are stable enough to reach the stratosphere, where UV releases their chlorine. Each chlorine atom then destroys ozone **catalytically**: it converts an ozone molecule to oxygen, is itself regenerated, and repeats the cycle many thousands of times before being removed. The same chemistry, intensified over the winter Antarctic by polar stratospheric clouds, produced the seasonal "ozone hole". Consequences of increased UV-B include skin cancers and cataracts, immune suppression, and damage to crops and aquatic phytoplankton — the base of marine food chains. The **Montreal Protocol** (1987) phased out CFCs and related substances worldwide and is the outstanding success story of international environmental law; the layer is slowly recovering, but because the chemicals persist for decades, recovery is measured in generations. Textile relevance: refrigeration and air-conditioning plant, foam and aerosol products in a factory all once used ozone-depleting substances, and compliant replacements are now a purchasing rule, not a preference.

Glossary

Pollution — introduction of substances or energy that harm organisms, materials or environmental uses. **Pollutant** — the substance or energy form causing pollution. **Primary / secondary pollutant** — emitted directly / formed by reaction in the environment. **Point / non-point source** — identifiable discharge point / diffuse input. **Temperature inversion** — warm air overlying cool air, suppressing vertical mixing. **Greenhouse effect** — warming of the surface by gases absorbing outgoing infrared radiation. **Global warming potential (GWP)** — warming potency of a gas relative to CO_2 over a stated period. **Photochemical smog** — oxidant mixture (ozone, PAN) formed by NO_x + VOCs in sunlight. **Ozone layer** — stratospheric region (~15–35 km) rich in ozone, absorbing UV-B. **Biomagnification** — increasing concentration of a persistent pollutant up a food chain.

Key points

• Pollution is a matter of concentration, location and rate — primary vs secondary, point vs non-point and biodegradable vs persistent are the working classifications.
• Nearly all air pollution lives in the troposphere; inversions trap it; the stratosphere holds the ozone layer and any pollutant reaching it persists for years.
• The greenhouse effect is natural and necessary; the problem is its enhancement by CO_2 (GWP 1), methane (≈ 28), nitrous oxide (≈ 265) and other gases.
• Photochemical smog = NO_x + VOCs + sunlight → ozone and PAN; cut either precursor to control it.
• CFC chlorine destroys ozone catalytically; the Montreal Protocol phased the chemicals out and the layer is slowly recovering.

Practice

1. Classify each as primary or secondary, and as point or non-point: (a) boiler soot from a chimney; (b) ground-level ozone over a city; (c) fertiliser run-off from farmland; (d) dyehouse effluent from an outlet pipe.
2. Explain how a temperature inversion turns a normal emission into a pollution episode.
3. A process emits 3,000 t CO_2, 20 t CH_4 and 4 t N_2O in a year. Using GWP values 1, 28 and 265, compute total emissions in tonnes CO_2e and state which gas contributes most after CO_2.
4. Write the sequence of reactions by which NO_x and VOCs produce photochemical smog, and name two control measures aimed at the precursors.
5. Why does one chlorine atom destroy many ozone molecules, and why will the ozone layer take decades to recover even though CFC production has stopped?

Answer key

1. (a) Primary, point. (b) Secondary, non-point (formed in air over the whole city). (c) Primary pollutants (nutrients), non-point. (d) Primary, point.
2. Normally air cools with height and rising warm polluted air mixes and dilutes. In an inversion a warm layer sits above cool surface air; the polluted air below cannot rise through it, vertical mixing nearly stops, and emissions accumulate at breathing level until concentrations reach harmful values.
3. CO_2: 3,000 × 1 = 3,000. CH_4: 20 × 28 = 560. N_2O: 4 × 265 = 1,060. Total = 3,000 + 560 + 1,060 = **4,620 tonnes CO_2e**. After CO_2, nitrous oxide contributes most (1,060 t CO_2e) despite the smallest mass emitted.
4. Sunlight photolyses NO_2 → NO + O; O + O_2 → O_3 (ozone). VOC oxidation produces peroxy radicals that convert NO back to NO_2 without consuming ozone, so ozone (and PAN) accumulate.

Controls: reduce NO_x (burner tuning/low-NO_x burners, vehicle exhaust catalysts) and reduce VOCs (solvent substitution, vapour recovery, leak control) — any two.

5. Chlorine acts as a catalyst: $Cl + O_3 \rightarrow ClO + O_2$, then $ClO + O \rightarrow Cl + O_2$ — the chlorine is regenerated each cycle and repeats it thousands of times before removal. Recovery is slow because CFCs already released persist in the atmosphere for decades and keep supplying chlorine to the stratosphere long after production ceased.
